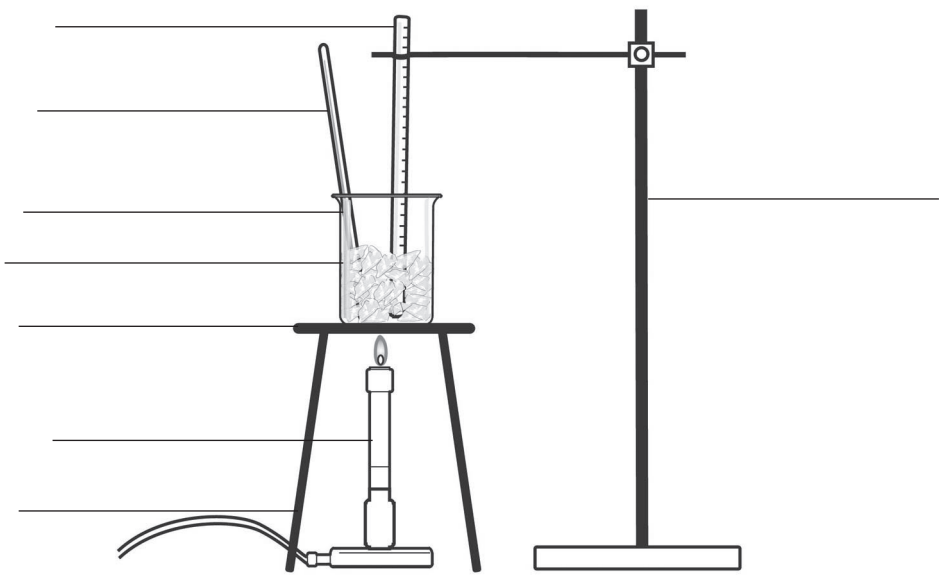
Temperature and state (1)

Poppy conducted an experiment to observe the temperature changes of crushed ice as it is heated. The scientific diagram below shows how her experiment was set up.

☞ **1** Use the word bank to label the equipment used.



Word bank

beaker

tripod

retort stand and clamp

stirring rod

Bunsen burner

crushed ice

thermometer

gauze mat

2 Below is a list of the steps that Poppy followed. Number the steps in the order in which they were completed.

_____ Record the temperature every minute.

_____ Place thermometer in ice without touching the bottom of the beaker.

_____ Set up the tripod, gauze mat and Bunsen burner on a heatproof mat.

_____ Use the stirring rod to constantly stir the ice slowly.

_____ Record the temperature of the ice before lighting the Bunsen burner.

_____ Place beaker on gauze mat.

_____ Light the Bunsen burner and heat on a medium flame.

_____ Half fill the beaker with crushed ice.

_____ When the water begins to boil, continue heating for another 3 minutes.

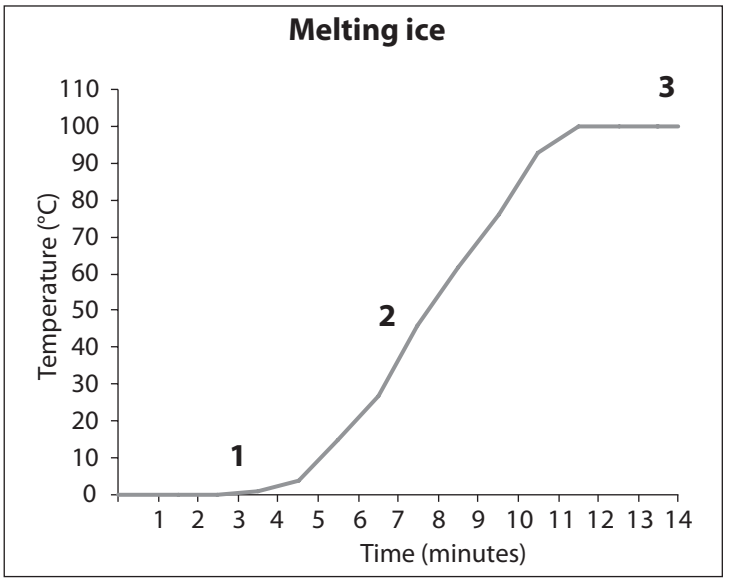
3 Why do you think that Poppy made sure the thermometer didn't touch the bottom of the beaker? _____

_____

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Temperature and state (2)

Poppy used her results to plot a graph showing temperature against time. This is called a heating curve.



☞ **1** Use the words in the word bank to answer the questions below.

Word bank					
boils	solid	liquid	melting	evaporation	heating up

a What is the state of matter indicated at Stage 1? _____

b What do you think is happening at Stage 1? _____

c What do you think is happening at Stage 2? _____

d Stage 2 represents what state of matter? _____

e What happens to the liquid at 100°C? _____

f What change of state is happening at Stage 3? _____

g Circle the melting point and boiling point on the graph. Use blue for the melting point and red for the boiling point.

2 Write down one thing you learnt from Poppy's experiment.

_____

_____

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